Astroshop dependency and tier model

A public astronomy e-commerce storefront. Customers browse a product catalog, add items to a cart, and check out through a synchronous order pipeline. Traffic is synthetic, driven by the load generator; faults are injected through flagd. Telemetry is OpenTelemetry throughout, scraped by Prometheus and traced through Jaeger.

Tiers

Tier 0 is the customer-facing critical path: a failure degrades the storefront directly. Tier 1 is supporting synchronous work — a failure degrades a tier-0 service but the storefront stays up. Tier 2 is asynchronous back-office processing off the request path entirely; a failure delays ledger and fraud data, not orders.

Tier 0: cart, checkout, frontend, frontend-proxy, payment, product-catalog

Tier 1: ad, astronomy-db, currency, flagd, image-provider, kafka, otel-collector, quote, recommendation, shipping, valkey-cart

Tier 2: accounting, email, fraud-detection

Reading the graph

The table on the following page lists every service, its tier, its owning team, the services it calls, and its blast radius — the set of services that transitively depend on it and therefore degrade when it does. The blast radius is what turns "service X is down" into "and here is who will be paged for it". Services with an empty *calls* column are leaves: they have no upstream cause, so their runbooks have no "rule out upstream" step.

otel-collector and flagd are in the deployment but excluded from this model: every service depends on them, so including them collapses the graph. The health rollup in app/system-monitor makes the same exclusion for the same reason.

Fault ownership

catalog (PD-CATALOG): adFailure, adHighCpu, adManualGc, productCatalogFailure, recommendationCacheFailure

checkout (PD-CHECKOUT): cartFailure, emailMemoryLeak, failedReadinessProbe, intlShippingSlowdown, paymentFailure, paymentUnreachable

data (PD-DATA): kafkaQueueProblems

storefront (PD-STOREFRONT): imageSlowLoad

Service dependency table

Service	Tier	Team	Calls	Blast radius
accounting	2	data	kafka	—
ad	1	catalog	—	frontend, frontend-proxy
astronomy-db	1	platform-eng	—	product-catalog, checkout, frontend, recommendation, frontend-proxy
cart	0	checkout	valkey-cart	checkout, frontend, frontend-proxy
checkout	0	checkout	cart, currency, email, payment, product-catalog, shipping	frontend, frontend-proxy
currency	1	checkout	—	checkout, frontend, frontend-proxy
email	2	checkout	—	checkout, frontend, frontend-proxy
flagd	1	platform-eng	—	—
fraud-detection	2	data	kafka	—
frontend	0	storefront	ad, cart, checkout, currency, product-catalog, quote, recommendation, shipping, image-provider	frontend-proxy
frontend-proxy	0	storefront	frontend	—
image-provider	1	storefront	—	frontend, frontend-proxy
kafka	1	data	—	accounting, fraud-detection
otel-collector	1	platform-eng	—	—
payment	0	checkout	—	checkout, frontend, frontend-proxy
product-catalog	0	catalog	astronomy-db	checkout, frontend, recommendation, frontend-proxy
quote	1	catalog	—	frontend, shipping, checkout, frontend-proxy
recommendation	1	catalog	product-catalog	frontend, frontend-proxy
shipping	1	checkout	quote	checkout, frontend, frontend-proxy
valkey-cart	1	platform-eng	—	cart, checkout, frontend, frontend-proxy